



SKG14 IC8 Pacing & Emerging Technologies for ICM Engineers L2

13.0 Introduction to 5G Wireless

Presentation to XXX

July 2025

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Learning Outcome

At the end of the session, you will be able to:

- Describe the concept of Industrial Wireless and 5G communication, including its architecture, components, and key differences from 4G.
- Explain the various spectrum allocations, limitations, and the differences between public and private 5G networks, including the 3.5GHz band.
- Develop an understanding of 5G applications and use cases in industrial environments while aligning with relevant standards and references.

13.0 Introduction to 5G

13.1 5G Technology Introduction

13.2 Differences between 4G and 5G

13.3 5G Architecture and Components

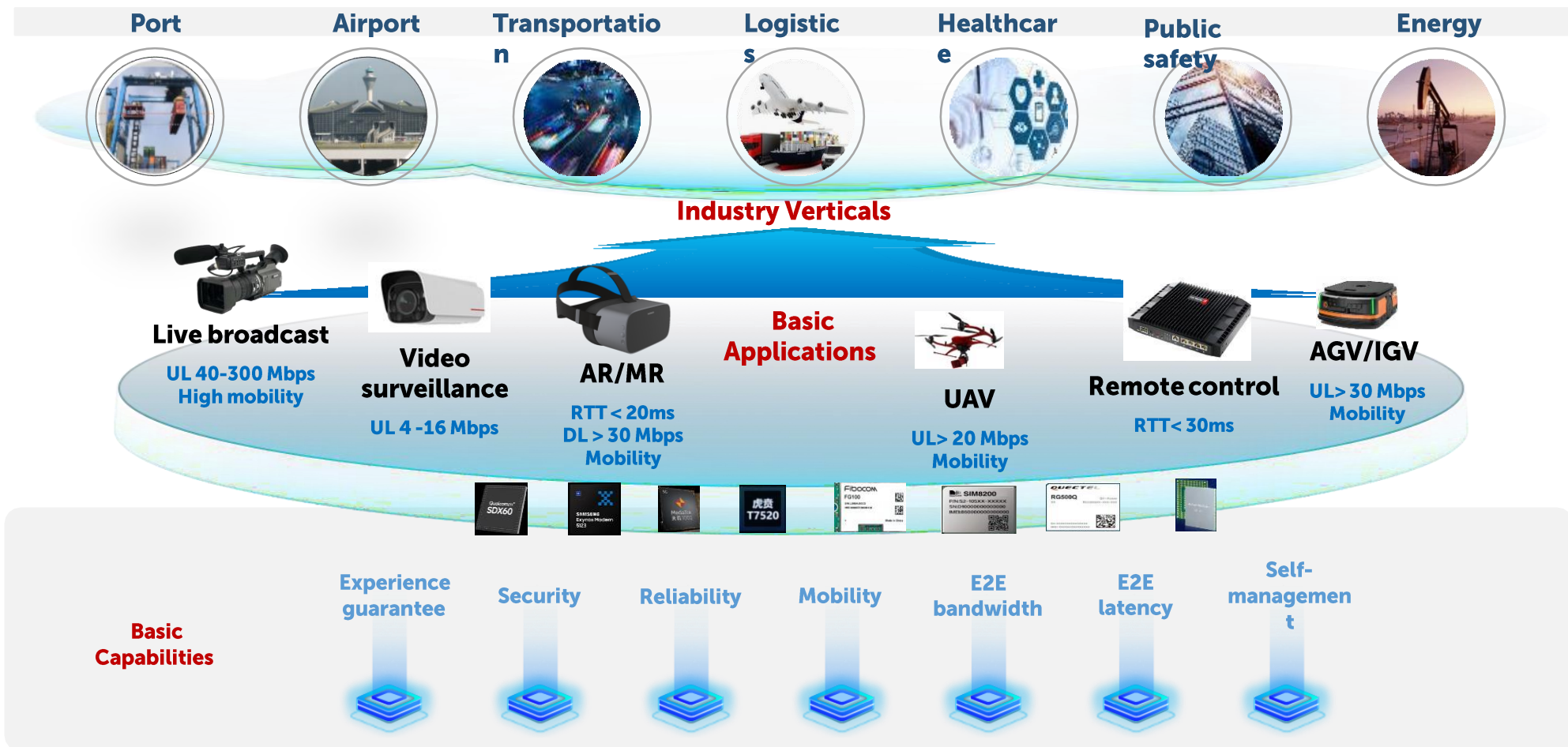
13.4 Differences between public and private 5G (3.5GHz for PETRONAS)

13.5 Various Spectrum Allocation and Limitation

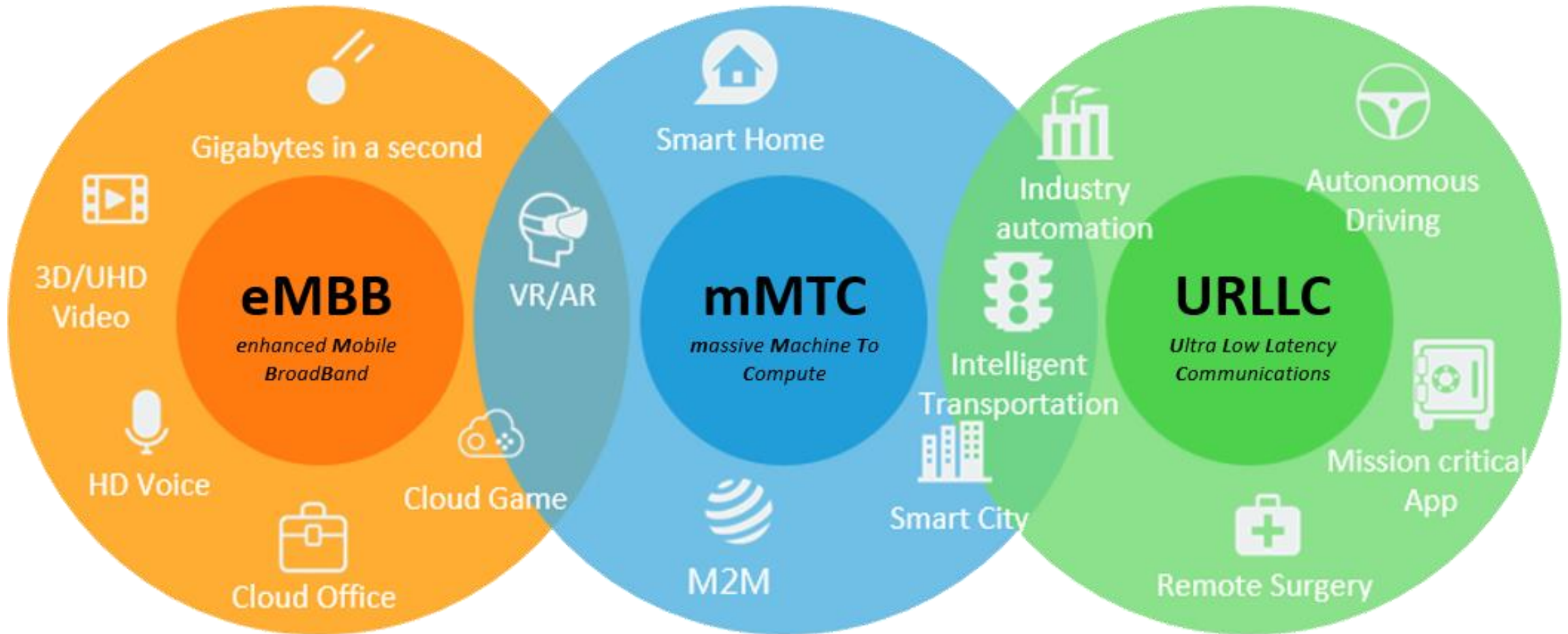
13.6 5G Application & Use Cases

13.7 5G References and Standards

5G Capabilities Enable Verticals Digital Transformation



13.1 5G Technology Introduction – 3 Attributes



13.2 Differences between 4G and 5G

The Evolution of 5G



1980s

1G

Analog Telecommunications



1990s

2G

Text Messaging



2000s

3G

Mobile and Wireless Internet Connection



2010s

4G

Cloud, IP and Mobile Broadband



2020s

5G

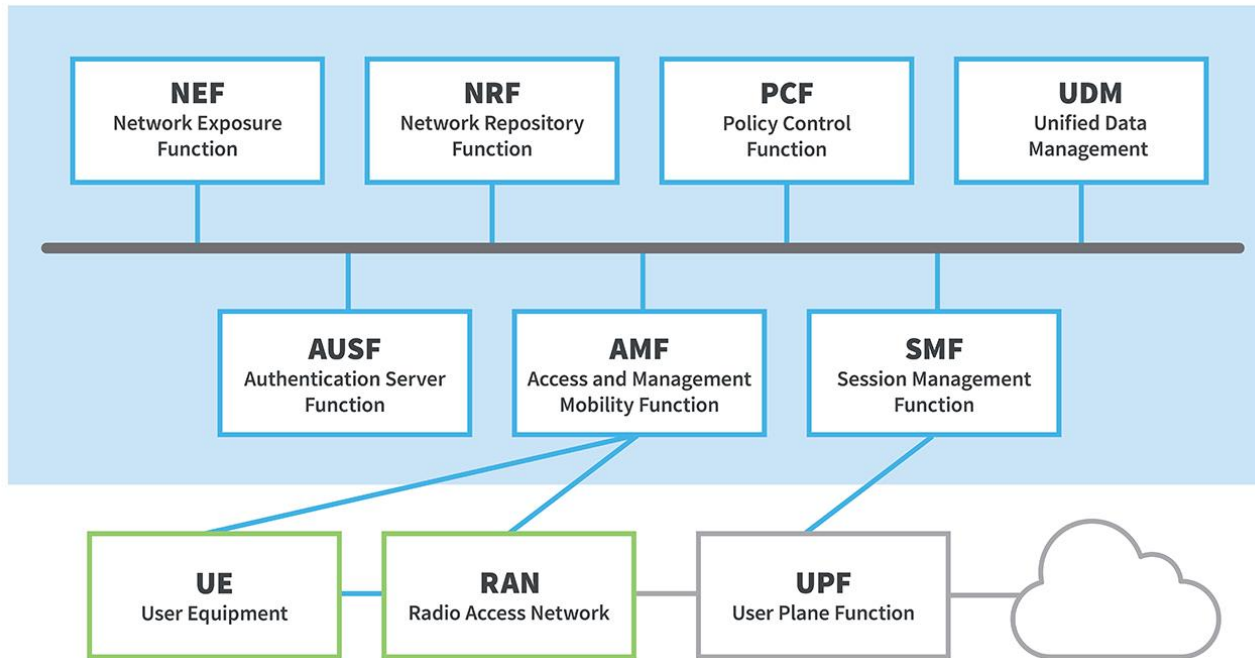
FASTER EVERYTHING!
Unlimited Data Capacity

Benefits of 5G over 4G

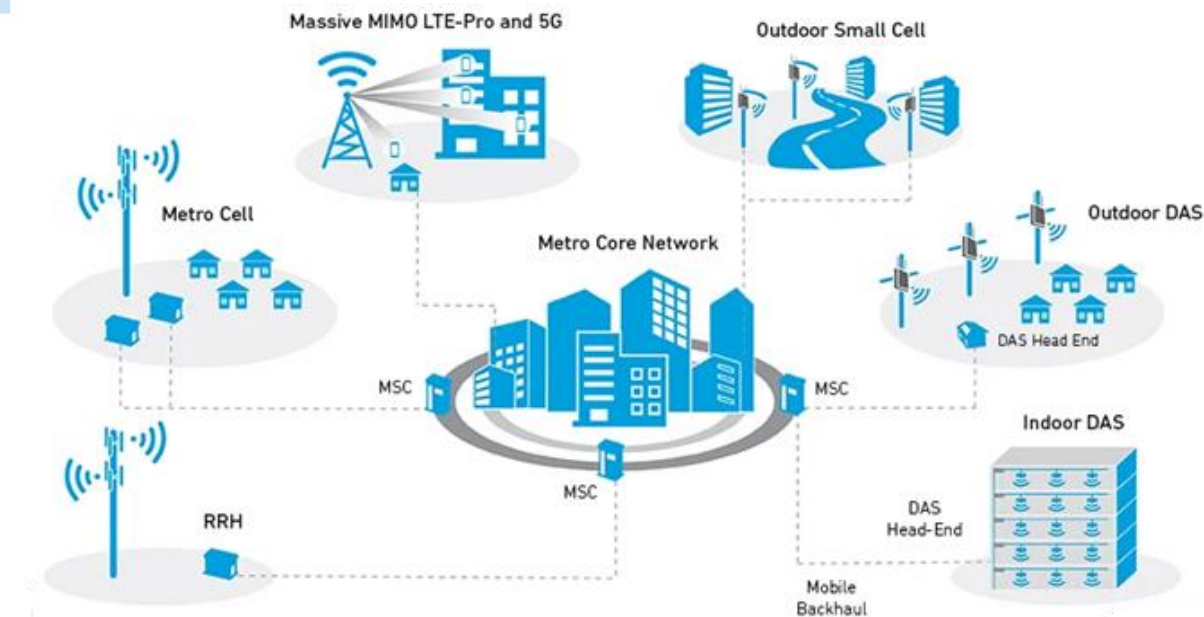
4G		5G
60-90 ms	01	01 Less than 1 ms
		Latency
1 Gbps	02	02 10 Gbps
		Potential download speed
Cell towers	03	03 Small Cells
		Base stations
20 MHz channels	04	04 100 MHz to 800 MHz channels
		OFDM Encoding
200 to 400 users per cell	05	05 100 times greater than 4G
		Goal for cell density

Source: <https://www.earthlink.net/blog/what-is-5g/>

13.3 5G Architecture and Components



Source: <https://www.sdxcentral.com/5g/definitions/key-elements-5g-network/>



Source <https://www.digi.com/blog/post/5g-network-architecture>

13.3 5G Architecture and Components

Key Components of 5G Architecture

User Equipment (UE)

Radio Access Network (RAN)

Core Network (5GC)

Transport Network

Network Slicing

Edge Computing

13.3 5G Architecture and Components

Advantages

Performance

- High-speed, reliable, and low-latency connectivity.

Scalability

- Supports billions of connected devices.

Flexibility

- Adapts to diverse industries like oil and gas, healthcare, manufacturing, and transportation.

13.3 5G Architecture and Components

5G Network Cloud

Purpose of 5G Network Cloud:

- Virtualized, cloud-native infrastructure to enable scalable, flexible, and efficient network operations.
- Supports diverse industries with tailored solutions
- Centralized control support which can disseminate directly to 5G edge networks.

13.3 5G Network Cloud Features

Network Slicing

Edge Computing

Flexibility

Scalability

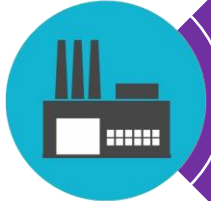
13.3 5G Network Cloud

Benefit for Industries



Healthcare

- Remote surgery, patient monitoring, and telemedicine powered by ultra-low latency.



Manufacturing

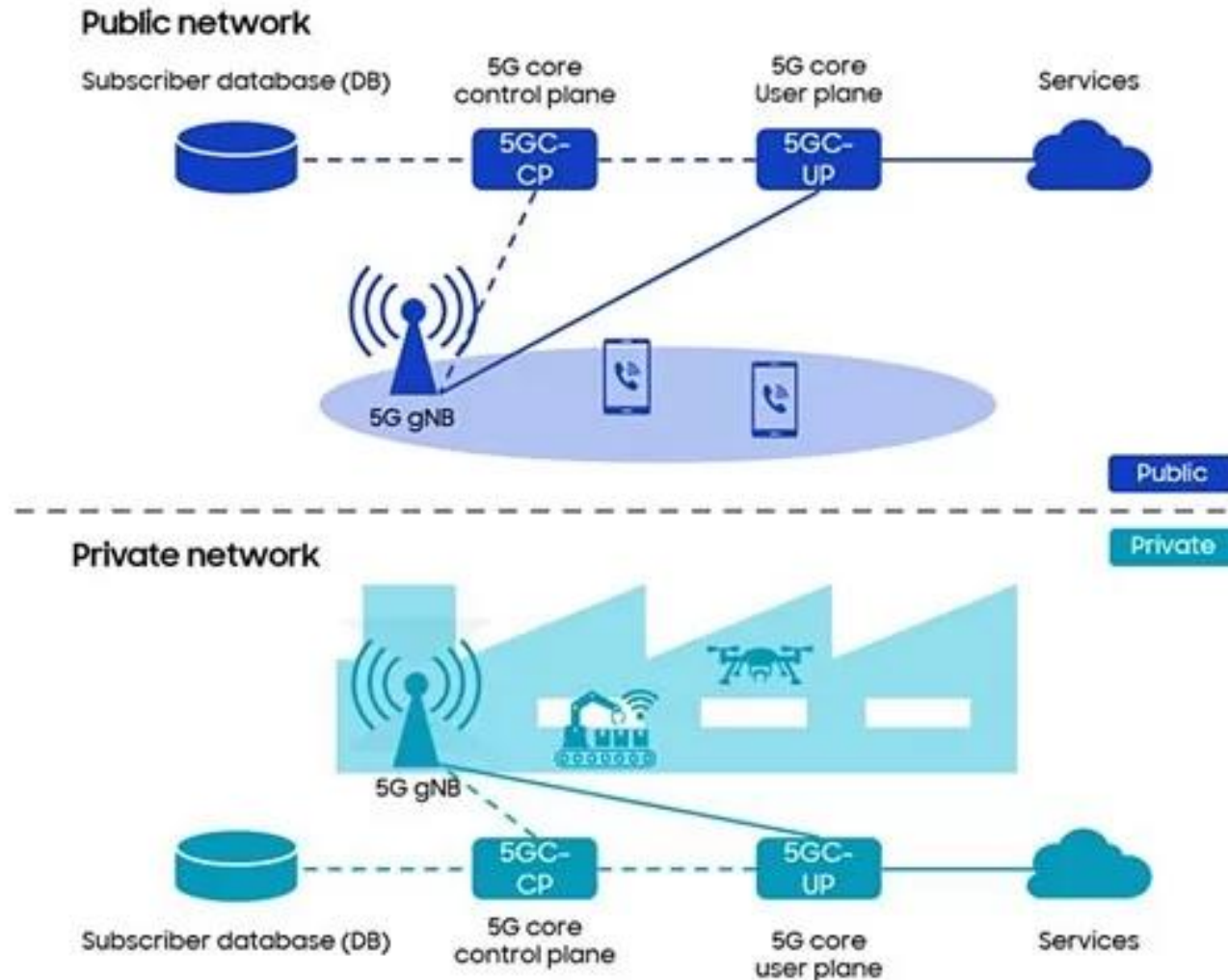
- Real-time automation, predictive maintenance, and robotic control for Industry 4.0.



Transportation

- Supports connected vehicles, traffic management, and autonomous driving.

13.4 Differences between public and private 5G (3.5GHz for PETRONAS)



Source: <https://5ghub.us/5g-private-networks-architecture-use-cases-challenges-and-solutions/>

13.5 5G Frequency Band in Malaysia



Source: <https://www.mcmc.gov.my/en/media/press-releases/final-report-on-allocation-of-spectrum-bands-for-m>

13.5 Various Spectrum Allocation and Limitation

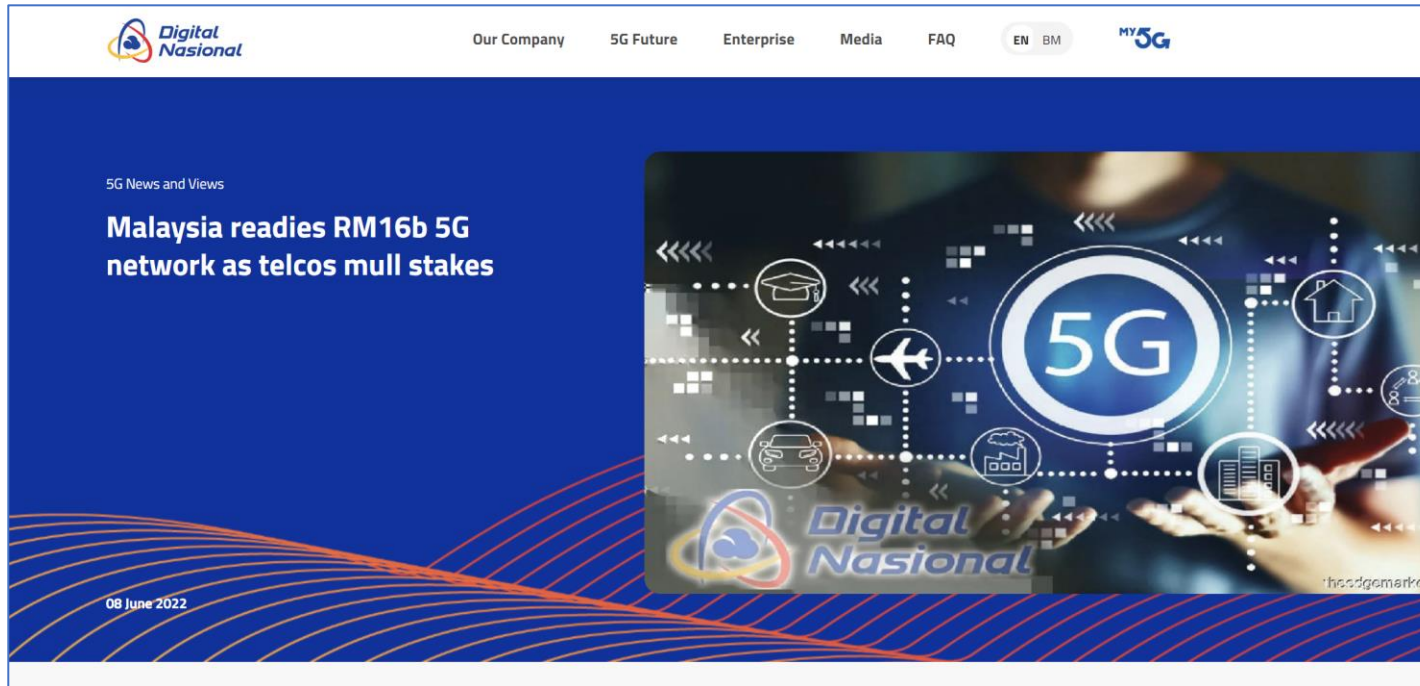
5G roll out is part of Malaysia Digital Economy Blueprint launched in February 2021.

Frequency Band	Allocation	Characteristics	Limitations
700 MHz	Assigned to Digital Nasional Berhad (DNB)	Wide coverage, good indoor penetration, suitable for rural areas	Lower data rates compared to higher frequency bands
3.5 GHz	Allocated to DNB for 5G deployment	Balanced coverage and capacity, suitable for urban areas	Moderate coverage, requires denser base stations
26/28 GHz	Identified by MCMC for initial 5G rollout, assigned via tender & for private networks	Ultra-high data rates, ideal for industrial automation and HD video streaming	Limited range, poor building penetration, needs dense small cells

13.5 Various Spectrum Allocation and Limitation

Challenges and Considerations:

- Spectrum Sufficiency
- Deployment Strategy



Govt picks U Mobile to implement Malaysia's second 5G network

By Digital News Asia November 1, 2024

- MCMC said it ran rigorous technical and commercial processes on all bids
- With MCMC nod, U Mobile can collaborate with other MNOs in implementation



Source: <https://mcmc.gov.my/en/spectrum/spectrum-management>

13.6 5G Application & Use Cases

5G coverage provide connectivity various native 5G Use Cases such as 5G Ex-rated smartphones, drone, robotics, and CCTV.



Source: <https://dgtlinfra.com/5g-use-cases-in-10-different-industries/>

13.7 5G References and Standards

Reference:

- The Communications and Multimedia Act 1998 and the Communications and Multimedia (Spectrum) Regulations 2000 are the legal frameworks that govern 5G licensing in Malaysia.
- The Malaysian Communications and Multimedia Commission (MCMC) is the authority responsible for managing and regulating the use of the radio spectrum in Malaysia.
- The MCMC has deployed a dual wholesale network for 5G services that includes Digital Nasional Berhad and UMobile as network providers.
- PETRONAS 5G Reference No: PTG 14.40.41 PETRONAS 5G Network System. The objective of this document is to provide 5G technical guidelines covering design, engineering and its implementation for entire PETRONAS and its subsidiaries.

Standard:

- 3GPP is the organization responsible for developing the 5G standard.
- **3GPP Release 15** introduced the first phase of 5G standard, including a new radio transmission technique and key concepts like enhanced mobile broadband (eMBB), ultra-reliable low latency communications (URLLC), and massive machine-type communications (mMTC).
- Subsequent releases, such as Release 16 and Release 17, have continued to enhance and expand the capabilities of 5G.

Source: [Malaysian Communications And Multimedia Commission \(MCMC\) | Suruhanjaya Komunikasi dan Multimedia Malaysia \(SKMM\) - Spectrum Management](#)

